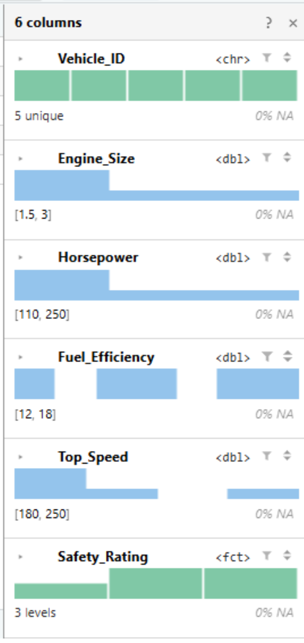
SET 28. Vehicle Performance Analysis

Dataset:

	Vehicle_ID	Engine_Size	Horsepower	Fuel_Efficiency	Top_Speed	Safety_Rating
1	V1	1.5	110	18	180	4
2	V2	2.0	150	15	200	5
3	V3	3.0	250	12	250	5
4	V4	2.5	200	14	220	4
5	V5	1.8	130	17	190	3

Showing 1 to 5 of 5 entries, 6 total columns



1. Using R, develop a violin plot to compare the distribution of Fuel_Efficiency across different Safety_Rating categories. Examine the spread, central tendency, and variability within each group, and interpret what this indicates about efficiency patterns.

2. Using R, construct a scatter plot to explore the relationship between Horsepower and Top_Speed, applying color differentiation based on Engine_Size. Analyze the strength and direction of correlation and find observable performance trends.

3. Using R, create a correlation heatmap for all numerical variables in the dataset. Identify the variables most strongly associated with Top_Speed and their influence on overall vehicle performance.

4. In Tableau, design an interactive dashboard incorporating bar charts and scatter plots, along with filters for Engine_Size and Safety_Rating, to highlight vehicles that balance fuel efficiency and high performance. Provide analytical interpretation of the insights derived from the dashboard.

R Program (Dataset Included)

Copy and run this code in RStudio.

```
# ==========================================================

# SET 28 - VEHICLE PERFORMANCE ANALYSIS
```

```
# ============================================================

# Load package
library(ggplot2)

# -----------------------------
# Dataset
# -----------------------------
vehicle <- data.frame(
Vehicle_ID = c("V1","V2","V3","V4","V5"),
Engine_Size = c(1.5,2.0,3.0,2.5,1.8),
Horsepower = c(110,150,250,200,130),
Fuel_Efficiency = c(18,15,12,14,17),
Top_Speed = c(180,200,250,220,190),
Safety_Rating = factor(c(4,5,5,4,3))
)

print(vehicle)
View(vehicle)

# ============================================================
# 1. Violin Plot
# ============================================================

ggplot(vehicle,
aes(x = Safety_Rating,
y = Fuel_Efficiency,
fill = Safety_Rating)) +
geom_violin(trim = FALSE, alpha = 0.7) +
geom_jitter(width = 0.1, size = 2) +
labs(
title = "Fuel Efficiency by Safety Rating",
```

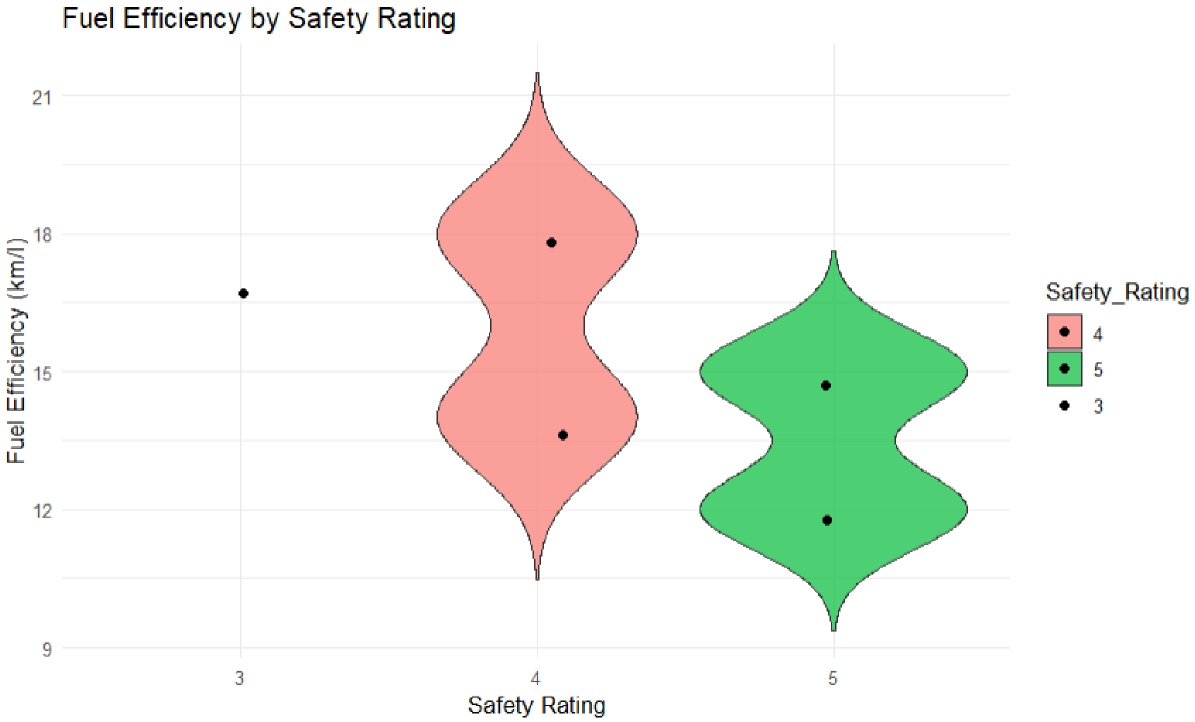
```
x = "Safety Rating",
y = "Fuel Efficiency (km/l)"
) +
theme_minimal()

# ============================================================
# 2. Scatter Plot
# ============================================================

ggplot(vehicle,
aes(x = Horsepower,
y = Top_Speed,
color = Engine_Size)) +
geom_point(size = 4) +
geom_smooth(method = "lm", se = FALSE, color = "black") +
labs(
title = "Horsepower vs Top Speed",
x = "Horsepower",
y = "Top Speed (km/h)",
color = "Engine Size"
) +
theme_minimal()

# ============================================================
# 3. Correlation Heatmap
# ============================================================

num_data <- vehicle[, c(
"Engine_Size",
"Horsepower",
"Fuel_Efficiency",
"Top_Speed"
```

```
)]

corr_matrix <- cor(num_data)

print(corr_matrix)

heatmap(
corr_matrix,
Rowv = NA,
Colv = NA,
scale = "none",
col = colorRampPalette(c("blue","white","red"))(50),
margins = c(7,7),
main = "Correlation Heatmap"
)
# Display correlation values
round(corr_matrix, 2)
```

1. Violin Plot

```
# ============================================================
# SET 28 - VEHICLE PERFORMANCE ANALYSIS
# ============================================================
# Load package
library(ggplot2)
# ------------------------------
# Dataset
# ------------------------------
vehicle <- data.frame(
  Vehicle_ID = c("V1","V2","V3","V4","V5"),
  Engine_Size = c(1.5,2.0,3.0,2.5,1.8),
  Horsepower = c(110,150,250,200,130),
  Fuel_Efficiency = c(18,15,12,14,17),
  Top_Speed = c(180,200,250,220,190),
```

```
  Safety_Rating = factor(c(4,5,5,4,3))
)
print(vehicle)
View(vehicle)
# ============================================================
# 1. Violin Plot
# ============================================================
ggplot(vehicle,
       aes(x = Safety_Rating,
           y = Fuel_Efficiency,
           fill = Safety_Rating)) +
  geom_violin(trim = FALSE, alpha = 0.7) +
  geom_jitter(width = 0.1, size = 2) +
  labs(
    title = "Fuel Efficiency by Safety Rating",
    x = "Safety Rating",
    y = "Fuel Efficiency (km/l)"
  ) +
  theme_minimal()
```

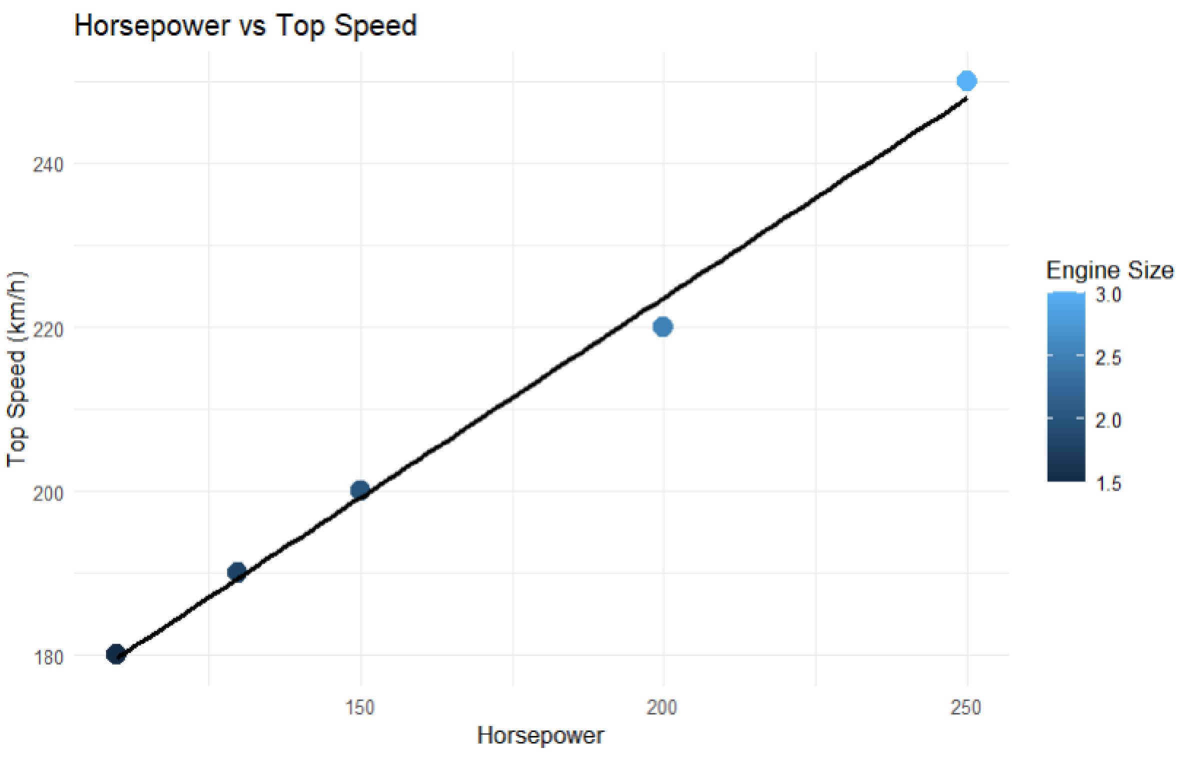


Output: Fuel Efficiency vs Safety Rating

Interpretation: The violin plot compares fuel efficiency across safety ratings. Vehicles with safety ratings 3 and 4 tend to have higher fuel efficiency, while rating 5 vehicles show lower efficiency because they have larger engines and higher performance.

2. Scatter Plot

```
ggplot(vehicle,
       aes(x = Horsepower,
           y = Top_Speed,
           color = Engine_Size)) +
  geom_point(size = 4) +
  geom_smooth(method = "lm", se = FALSE, color = "black") +
  labs(
    title = "Horsepower vs Top Speed",
    x = "Horsepower",
    y = "Top Speed (km/h)",
    color = "Engine Size"
  ) +
  theme_minimal()
```



Output: Horsepower vs Top Speed

Interpretation: There is a strong positive correlation between horsepower and top speed. As horsepower increases, top speed also increases. Larger engine sizes are associated with higher-performance vehicles.

3. Correlation Heatmap

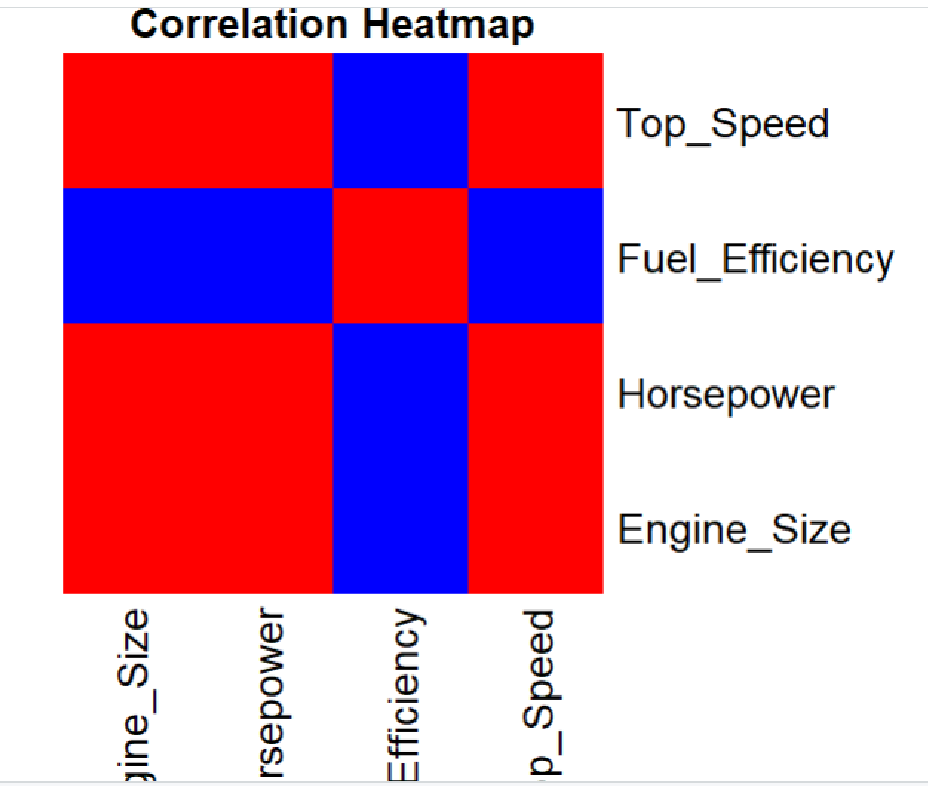
```
num_data <- vehicle[, c(
  "Engine_Size",
  "Horsepower",
  "Fuel_Efficiency",
  "Top_Speed"
)]
corr_matrix <- cor(num_data)
print(corr_matrix)
heatmap(
  corr_matrix,
  Rowv = NA,
  Colv = NA,
  scale = "none",
  col = colorRampPalette(c("blue","white","red"))(50),
  margins = c(7,7),
  main = "Correlation Heatmap"
)
```

The heatmap displays relationships among numerical variables.

Strongest relationships with Top Speed:

- Horsepower → Strong Positive
- Engine Size → Positive
- Fuel Efficiency → Negative

Interpretation: Higher horsepower and larger engines improve top speed, while fuel efficiency decreases as performance increases.



4. Tableau Dashboard

Sheet 1 – Bar Chart

- Columns → Vehicle_ID
- Rows → Fuel_Efficiency
- Color → Safety_Rating

Title: Fuel Efficiency by Vehicle

Sheet 2 – Scatter Plot

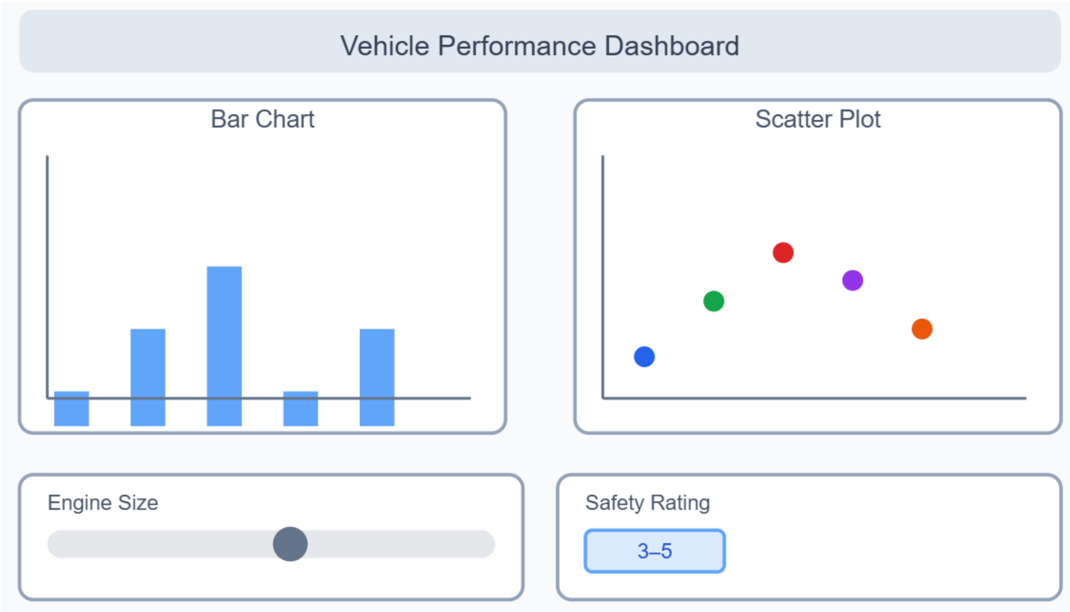
- Columns → Horsepower
- Rows → Top_Speed
- Color → Engine_Size
- Label → Vehicle_ID

Title: Horsepower vs Top Speed

Add Filters

- Engine_Size → Show Filter
- Safety_Rating → Show Filter

Dashboard Layout



Dashboard Interpretation

- V3 offers the highest horsepower and top speed but the lowest fuel efficiency.
- V1 and V5 provide better fuel efficiency with moderate performance.
- Filters help identify vehicles that balance high performance and good fuel economy, supporting better vehicle selection.

SET 29 .Student Academic Performance

Dataset:

	Student_ID	Age	Study_Hours	Attendance	Test_Score	Participation_Score
1	S1	19	12	90	85	8
2	S2	21	8	70	70	7
3	S3	20	15	95	92	9
4	S4	22	10	85	80	8
5	S5	23	7	60	65	6

1. Using R, create a stacked area chart showing Test_Score and Participation_Score across students. Analyze trends.

2. Using R, generate a boxplot of Study_Hours grouped by Attendance quartiles. Interpret patterns. Include axis labels, title, and colors for each quartile. Analyze the boxplot to identify median and interquartile range of study hours per attendance quartile and any outliers in study hours.

3. Using R, construct a density plot for Test_Score to evaluate score distribution and outliers.

4. In Tableau, design an interactive scatterplot with Study_Hours on X-axis, Test_Score on Y-axis, colored by Participation_Score. Include filters for Age.

```
# ============================================================
# SET 29 - STUDENT ACADEMIC PERFORMANCE
# ============================================================

library(ggplot2)
library(reshape2)

# Create dataset
student <- data.frame(
  Student_ID = c("S1","S2","S3","S4","S5"),
  Age = c(19,21,20,22,23),
  Study_Hours = c(12,8,15,10,7),
  Attendance = c(90,70,95,85,60),
  Test_Score = c(85,70,92,80,65),
  Participation_Score = c(8,7,9,8,6)
)

print(student)
View(student)


# ============================================================
# 1. STACKED AREA CHART
# Test Score and Participation Score across Students
# ============================================================

area_data <- melt(
  student[, c("Student_ID",
              "Test_Score",
```

```
            "Participation_Score")],
  id.vars = "Student_ID",
  variable.name = "Score_Type",
  value.name = "Score"
)

ggplot(
  area_data,
  aes(
    x = Student_ID,
    y = Score,
    fill = Score_Type,
    group = Score_Type
  )
) +
  geom_area(alpha = 0.7) +
  labs(
    title = "Test Score and Participation Score Across Students",
    x = "Student",
    y = "Score",
    fill = "Score Type"
  ) +
  theme_minimal()


# ============================================================
# 2. BOXPLOT OF STUDY HOURS BY ATTENDANCE QUARTILE
# ============================================================

# Create attendance quartiles
student$Attendance_Quartile <- cut(
  student$Attendance,
```

```
  breaks = quantile(
    student$Attendance,
    probs = c(0, 0.25, 0.50, 0.75, 1),
    na.rm = TRUE
  ),
  include.lowest = TRUE,
  labels = c(
    "Q1 - Low",
    "Q2",
    "Q3",
    "Q4 - High"
  )
)

# Boxplot
ggplot(
  student,
  aes(
    x = Attendance_Quartile,
    y = Study_Hours,
    fill = Attendance_Quartile
  )
) +
  geom_boxplot() +
  geom_jitter(
    width = 0.1,
    size = 3
  ) +
  labs(
    title = "Study Hours by Attendance Quartile",
    x = "Attendance Quartile",
    y = "Study Hours"
```

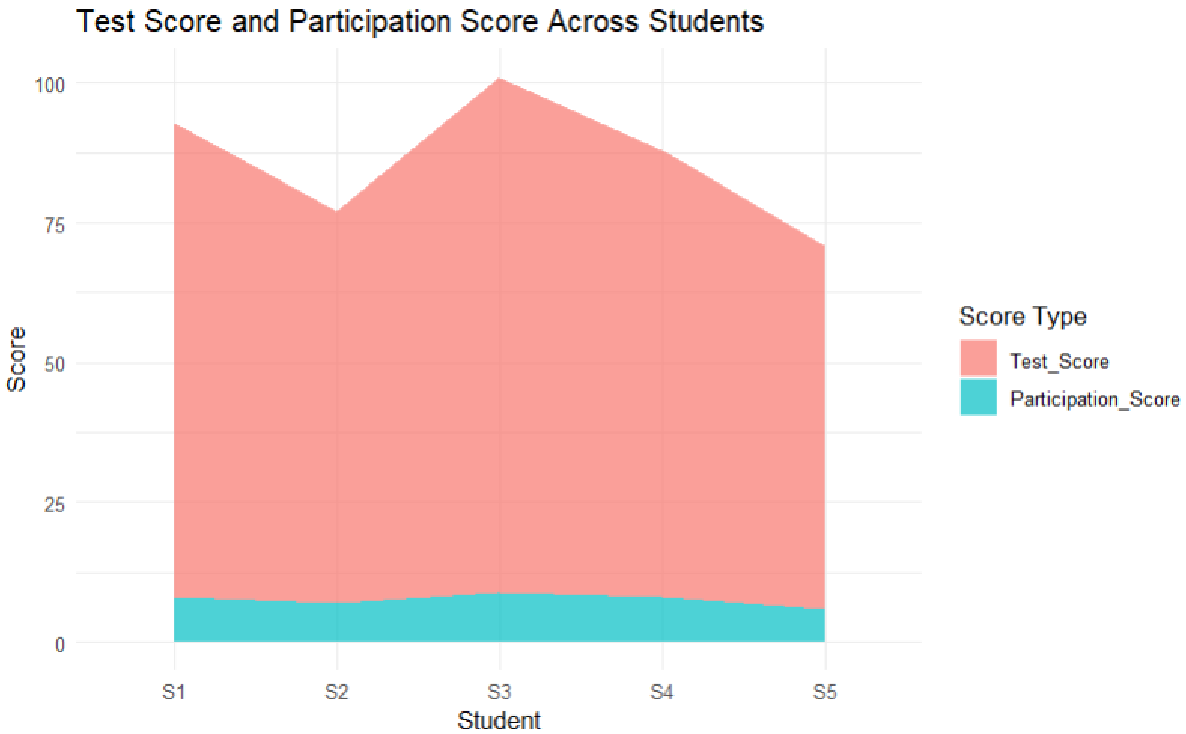
```
  ) +
  theme_minimal()


# ============================================================
# 3. DENSITY PLOT FOR TEST SCORE
# ============================================================

ggplot(
  student,
  aes(x = Test_Score)
) +
  geom_density(
    fill = "skyblue",
    alpha = 0.6
  ) +
  geom_rug() +
  labs(
    title = "Density Distribution of Test Scores",
    x = "Test Score",
    y = "Density"
  ) +
  theme_minimal()


# ============================================================
# OPTIONAL: IDENTIFY OUTLIERS USING IQR
# ============================================================

Q1 <- quantile(student$Study_Hours, 0.25)
Q3 <- quantile(student$Study_Hours, 0.75)
```

```
IQR_value <- Q3 - Q1

lower <- Q1 - 1.5 * IQR_value
upper <- Q3 + 1.5 * IQR_value

outliers <- student[
  student$Study_Hours < lower |
    student$Study_Hours > upper,
]

print(outliers)



# ==========================================================
# FINAL TABLE
# ==========================================================

print(student)
View(student)
```

1. Stacked Area Chart – Interpretation

```
area_data <- melt(
  student[, c("Student_ID",
              "Test_Score",
              "Participation_Score")],
  id.vars = "Student_ID",
  variable.name = "Score_Type",
  value.name = "Score"
)

ggplot(
  area_data,
```

```
  aes(
    x = Student_ID,
    y = Score,
    fill = Score_Type,
    group = Score_Type
  )
) +
  geom_area(alpha = 0.7) +
  labs(
    title = "Test Score and Participation Score Across Students",
    x = "Student",
    y = "Score",
    fill = "Score Type"
  ) +
  theme_minimal()
```

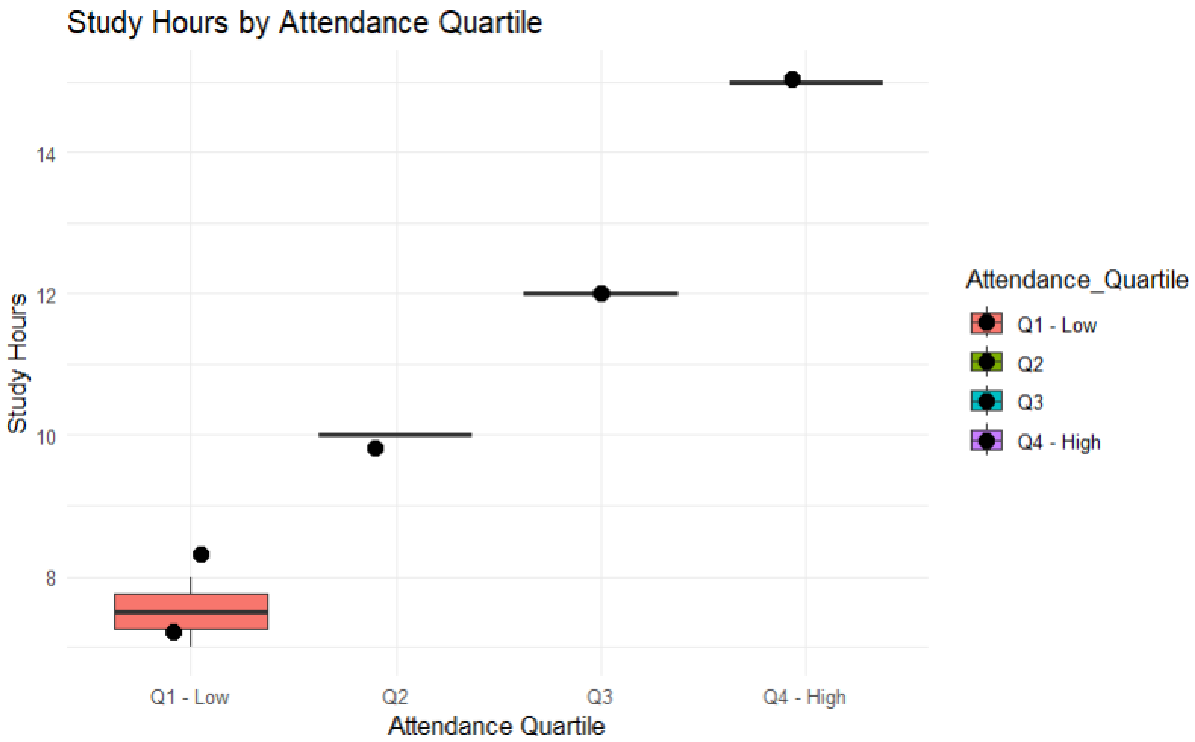


The chart compares **Test Score** and **Participation Score** across students. S3 has the highest Test Score (92) and Participation Score (9), while S5 has the lowest values. Overall, better participation is associated with better academic performance in this small dataset.

2. Boxplot – Interpretation

```
# Create attendance quartiles
student$Attendance_Quartile <- cut(
  student$Attendance,
  breaks = quantile(
    student$Attendance,
    probs = c(0, 0.25, 0.50, 0.75, 1),
    na.rm = TRUE
  ),
  include.lowest = TRUE,
  labels = c(
    "Q1 - Low",
    "Q2",
    "Q3",
    "Q4 - High"
  )
)

# Boxplot
ggplot(
  student,
  aes(
    x = Attendance_Quartile,
    y = Study_Hours,
    fill = Attendance_Quartile
  )
) +
  geom_boxplot() +
  geom_jitter(
    width = 0.1,
    size = 3
```

```
) +
labs(
  title = "Study Hours by Attendance Quartile",
  x = "Attendance Quartile",
  y = "Study Hours"
) +
theme_minimal()
```

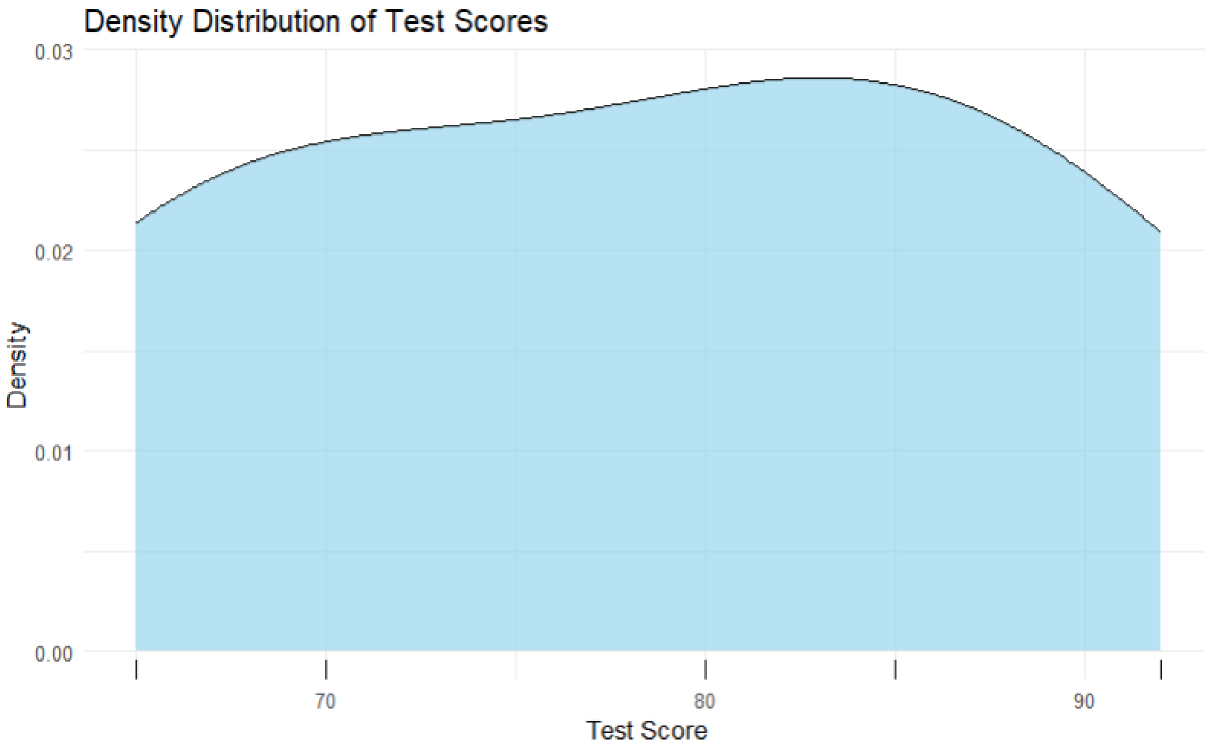


The boxplot compares Study Hours across Attendance quartiles. Students with higher attendance generally show higher study hours. The boxplot displays the **median, interquartile range (IQR), and possible outliers**. With only five students, the quartile groups are very small, so conclusions should be interpreted cautiously.

3. Density Plot – Interpretation

```
ggplot(
  student,
  aes(x = Test_Score)
) +
  geom_density(
    fill = "skyblue",
    alpha = 0.6
```

```
) +
geom_rug() +
labs(
  title = "Density Distribution of Test Scores",
  x = "Test Score",
  y = "Density"
) +
theme_minimal()
```



The density plot shows the distribution of Test Scores. Scores range from **65 to 92**, with most scores concentrated around the middle-to-high range. With only five observations, the density curve is only an approximate representation and cannot reliably establish outliers or the underlying population distribution.

---

4. Tableau Dashboard

Scatter Plot

In Tableau:

- Drag **Study_Hours** → Columns
- Drag **Test_Score** → Rows
- Drag **Participation_Score** → Color
- Drag **Student_ID** → Label

Title:

Study Hours vs Test Score

Age Filter

Drag:

Age → Filters → Show Filter

Dashboard

Create:

Dashboard → New Dashboard

Add the scatter plot and Age filter.

Suggested layout:

```
+-------------------------------------------+
|                                |
|      Study Hours vs Test Score            |
|                                |
|                                |
+-------------------------------------------+
| Age Filter                          |
+-------------------------------------------+
```

Dashboard Interpretation

Students with higher study hours generally have higher test scores. S3 has the highest study hours (15) and highest test score (92), while S5 has the lowest study hours (7) and test score (65). The Age filter allows users to explore whether these patterns differ across age groups.

SET 30

Dataset: Mobile_App_Usage.csv

1. Import the dataset in R. Plot a histogram and density plot for Screen_Time. Add suitable colors and labels. Interpret the screen-time distribution.

2. Create a scatter plot using R programming between Data_Used and Screen_Time. Calculate the correlation between them. Add a trend line. Explain the relationship.

3. Using R, find the average Satisfaction score for each Gender. Create a bar chart for the averages. Add value labels. Interpret user satisfaction patterns.

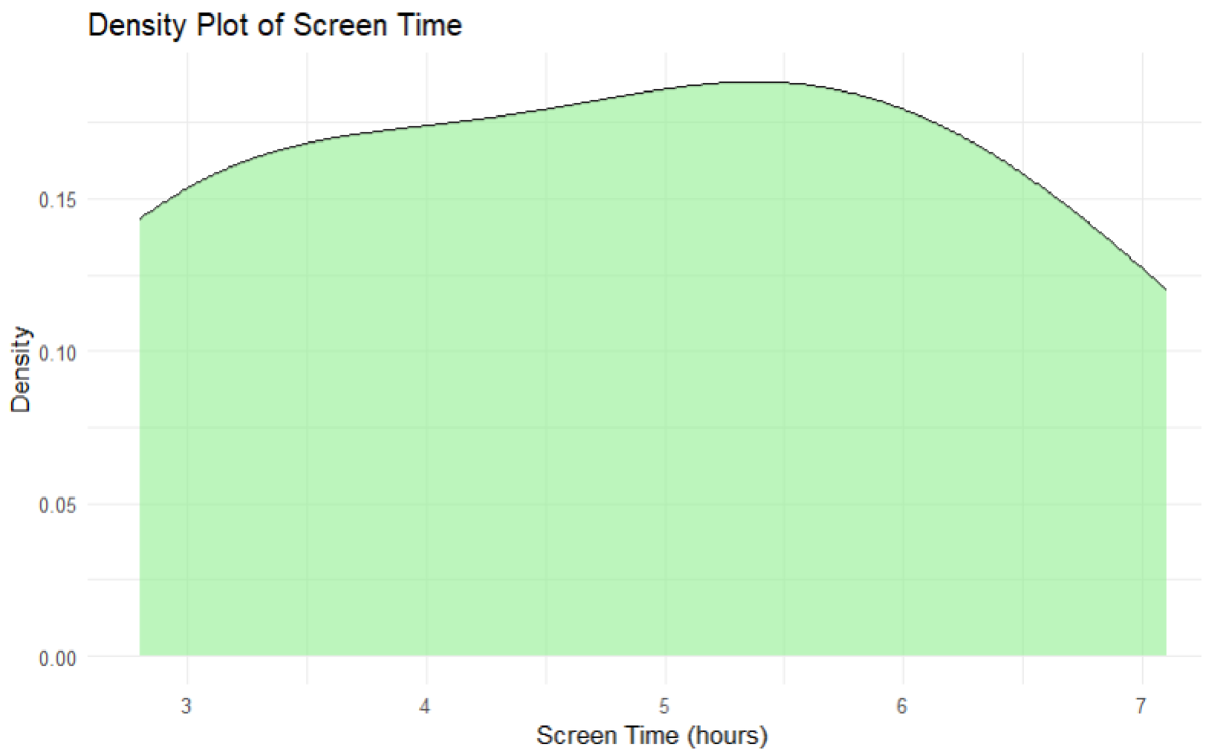
4. Import Mobile_App_Usage.csv into Tableau. Create a bubble chart (Screen Time vs Data Used). Use color for Gender and size for App Usage Count. Create a line chart for usage trends. Add filters for Gender and Usage_Date. Design a mini dashboard.

Mobile App Usage Analysis

1. Histogram and Density Plot for Screen Time

```
library(ggplot2)

# Create dataset directly in R
mobile <- data.frame(
  User_ID = c("U01","U02","U03","U04","U05","U06"),
  Gender = c("Male","Female","Male","Female","Male","Female"),
  Age = c(20,22,19,21,23,20),
  Screen_Time = c(4.5,6.0,3.2,7.1,2.8,5.4),
  App_Usage_Count = c(18,25,12,30,10,22),
  Data_Used = c(2.4,3.8,1.6,4.5,1.2,3.1),
  Satisfaction = c(3,5,3,5,2,4),
  Usage_Date = c(
    "2025-01-08","2025-01-08",
    "2025-02-11","2025-02-11",
    "2025-03-14","2025-03-14"
  )
)

# Convert date
mobile$Usage_Date <- as.Date(mobile$Usage_Date)

# Display dataset
print(mobile)
View(mobile)
```

```r
# Histogram
ggplot(mobile, aes(x = Screen_Time)) +
  geom_histogram(
    binwidth = 1,
    fill = "skyblue",
    color = "black"
  ) +
  labs(
    title = "Distribution of Screen Time",
    x = "Screen Time (hours)",
    y = "Number of Users"
  ) +
  theme_minimal()

# Density Plot
ggplot(mobile, aes(x = Screen_Time)) +
  geom_density(
    fill = "lightgreen",
    alpha = 0.6
  ) +
  labs(
    title = "Density Plot of Screen Time",
    x = "Screen Time (hours)",
    y = "Density"
  ) +
  theme_minimal()
```



Interpretation

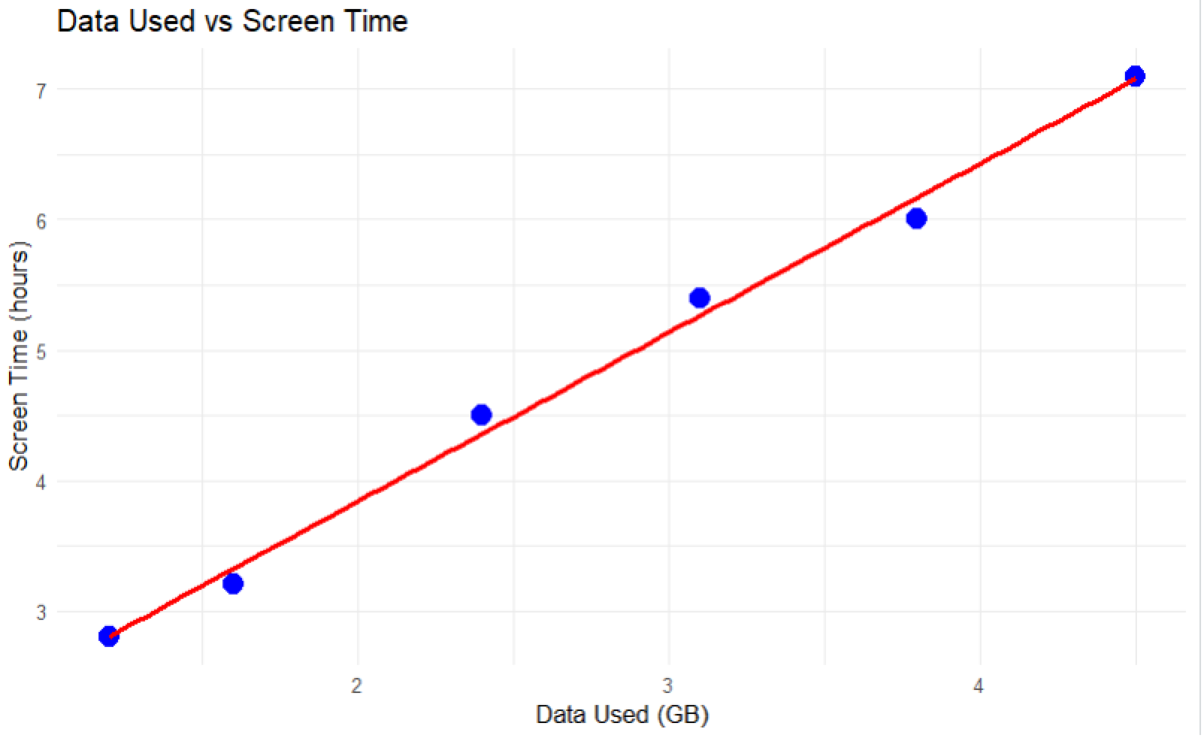
Screen time ranges from **2.8 to 7.1 hours**. The distribution shows that users have varying levels of mobile usage, with female users in this small dataset generally having higher screen time.

---

2. Scatter Plot – Data Used vs Screen Time

```
# Calculate correlation
correlation <- cor(
  mobile$Data_Used,
  mobile$Screen_Time
)

print(correlation)

# Scatter plot with trend line
ggplot(
  mobile,
  aes(
    x = Data_Used,
    y = Screen_Time
```

```
  )
) +
  geom_point(
    size = 4,
    color = "blue"
  ) +
  geom_smooth(
    method = "lm",
    se = FALSE,
    color = "red"
  ) +
  labs(
    title = "Data Used vs Screen Time",
    x = "Data Used (GB)",
    y = "Screen Time (hours)"
  ) +
  theme_minimal()
```



Interpretation

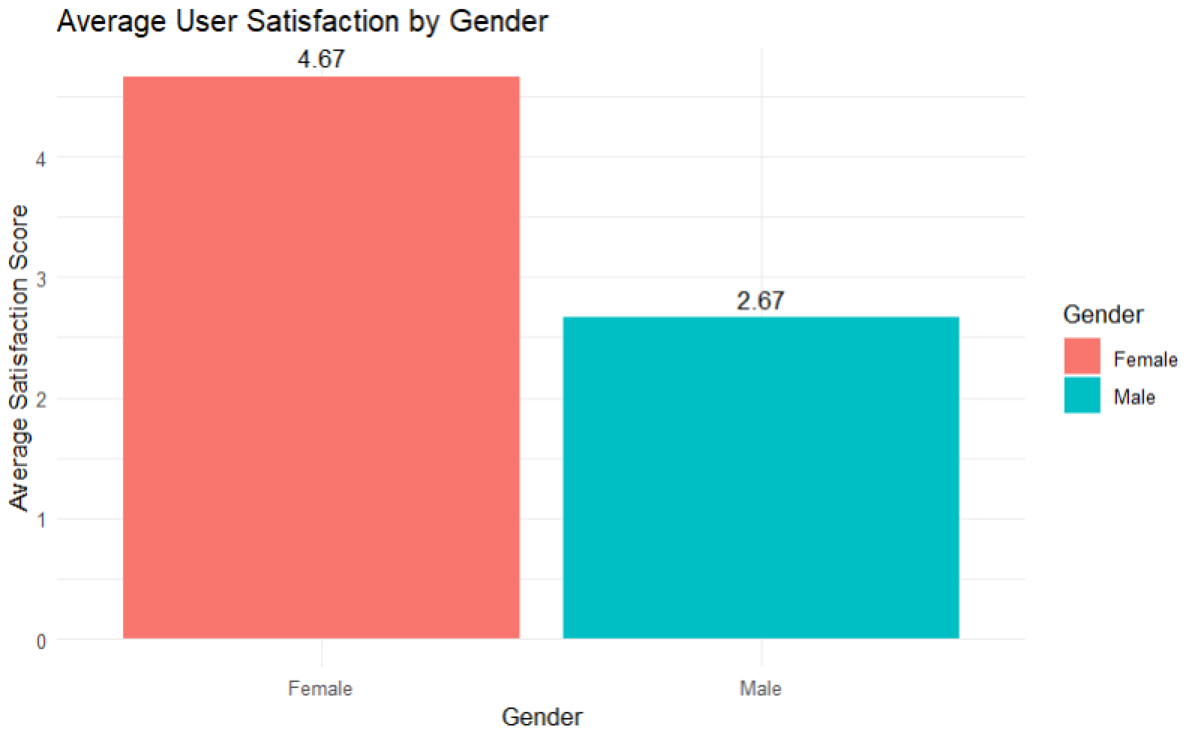
The correlation is **strong and positive** (approximately **0.99**). This indicates that users with higher screen time tend to consume more mobile data.

---

3. Average Satisfaction by Gender

```
# Calculate average satisfaction
avg_satisfaction <- aggregate(
  Satisfaction ~ Gender,
  data = mobile,
  FUN = mean
)

print(avg_satisfaction)

# Bar chart
ggplot(
  avg_satisfaction,
  aes(
    x = Gender,
    y = Satisfaction,
    fill = Gender
  )
) +
  geom_bar(
    stat = "identity"
  ) +
  geom_text(
    aes(
      label = round(Satisfaction, 2)
    ),
    vjust = -0.5
  ) +
  labs(
    title = "Average User Satisfaction by Gender",
```

```
  x = "Gender",
  y = "Average Satisfaction Score"
) +
theme_minimal()
```



Expected result

Gender	Average Satisfaction
Female	4.67
Male	2.67

Interpretation

Female users have a higher average satisfaction score than male users in this dataset. However, there are only three users in each group, so this result should not be generalized to a larger population.

---

4. Tableau Dashboard

Import Mobile_App_Usage.csv into Tableau.

Bubble Chart

Create a new worksheet.

- Screen_Time → Columns
- Data_Used → Rows
- Gender → Color

- App_Usage_Count → Size
- User_ID → Label

Title:

Screen Time vs Data Used

This creates bubbles where larger bubbles represent higher app usage counts.

Line Chart

Create another worksheet.

- Usage_Date → Columns
- Screen_Time → Rows
- Select **Line**

Title:

Mobile Usage Trend

Filters

Add:

- **Gender**
- **Usage_Date**

Select **Show Filter**.

Dashboard

Go to:

Dashboard → New Dashboard

Add:

1. Screen Time vs Data Used bubble chart
2. Mobile Usage Trend line chart
3. Gender filter
4. Usage Date filter

Suggested layout:

```
+----------------------------------------------+
|            MOBILE APP USAGE                  |
+----------------------------------------------+
|                                  |
|     Bubble Chart        Line Chart        |
|  Screen Time vs Data     Usage Trend          |
```

```
|                                    |
+------------------------------------------------+
| Gender Filter      Usage Date Filter      |
+------------------------------------------------+
```

Tableau Interpretation

The dashboard helps identify users with **high screen time, high data consumption, and frequent app usage**. Filtering by gender or usage date allows users to explore changing usage patterns interactively.